

Unexpected biotic substrata utilised by newly settled recruits of the sea cucumber *Cucumaria frondosa* in their nursery habitat

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Abstract

Recently, the occurrence of newly settled recruits and early juveniles of the sea cucumber *Cucumaria frondosa* was described in shallow waters (depth <2.5 m) of Qikiqtait, Nunavut (northern Canada) on shells of live and dead mussels, and on the bare surfaces of stones, forming a nursery habitat. It was assumed that most young individuals (i.e. <2 y) occurred on these substrata. However, since the initial publication by Hamel et al. (2023), new biotic substrata have been identified to host newly recruited juveniles in their nursery habitat, including the holdfast and blade of seaweeds, sea urchin spines, encrusting bryozoans, barnacles, coralline algae, and bivalves other than mussels, suggesting that the recruitment of *C. frondosa* may be more complex and opportunistic than originally suspected. As *C. frondosa* is currently the most heavily fished sea cucumber in the world, with fishing initiatives emerging in the Arctic, knowledge of where its recruits and juveniles are distributed is of major importance to protecting this potentially vulnerable resource.

Background

Despite the growing number of field reports that mentioned sightings of juveniles from various species of sea cucumbers (Wolfe et al. 2024), few studies have effectively detected new recruits (i.e. recently settled individuals) and identified nursery habitats (Mercier et al. 1999; Shiell 2004; Rogers et al. 2021; Hamel et al. 2023). Nevertheless, based on field observations, some sea cucumbers have been shown to have specific settlement and recruitment requirements (Mercier et al. 1999; Rogers et al. 2021; Hamel et al. 2023; Wolfe et al. 2024).

The dendrochirotid sea cucumber *Cucumaria frondosa* broadly distributed in the North Atlantic and Arctic oceans, and is commercially exploited in North America and Europe (Gianasi et al. 2021; Mercier et al. 2023). Small juveniles, 15–30 mm long, were documented on gravel and boulders, in shallow-water environments (<10 m deep) in the St Lawrence River Estuary (Hamel and Mercier 1996). However, recruits (i.e. <1–2 mm, and presumed to be <2 y old), have so far only been reported in Qikiqtait (Nunavut, northern Canada) on bare surfaces (i.e. visible absence of any macro-organisms) of stones and on shells of live and dead mussels (also on their byssus; *Mytilus* sp.) in waters between 1 m and 2.5 m deep (Hamel et al. 2023).

The present contribution summarises all other substrata on which annual recruits and early juveniles of *C. frondosa* were observed during additional sampling campaigns conducted over a two-year period in Qikiqtait. Specifically, sampling occurred in Kataaluk (56.603300°N, 79.150200°W) at a depth of ~1.0–3.5 m; Kataapik (56.629238°N, 79.269109°W) at a depth of ~1–3 m depth;

Katak (56.537183°N, 79.132764°W) at a depth of ~2–3 m; and east of the hamlet of Sanikiluaq (56.548755°N, 79.205297°W) in tide pools at low tide. Subtidal and intertidal collections of various substrata between 2023 and 2024 were analysed opportunistically. For each recruit and juvenile observed in the field (~1400 individuals examined), the substratum on which they were attached was recorded, with a focus on novel substrata.

Although new recruits of *C. frondosa* can reach densities upwards of 100 individuals/m² (Hamel et al. 2023), these levels were recorded almost exclusively on blue mussels (shell and byssus) and bare stones (generally on undersurfaces and in crevices). The present report documents a few exceptions to these substratum preferences by new recruits of the current year (i.e. estimated to be ≤2 mm in length). At least 30 recruits were found on the elongated blade of *Chorda* cf. *borealis* (~50 blades examined), themselves attached to a large stone (36 cm in diameter) at ~1 m depth (Table 1 and Fig. 1A, B). On the holdfast and blade of *Fucus distichus*, six recruits were found in tide pools and four at ~1 m depth (Table 1 and Fig. 1A, C). Three recruits were found attached to coralline algae (probably belonging to the genus *Lithothamnion*) at a depth of ~1–3 m (Table 1 and Fig. 1D). Additionally, a few dozen recruits were found settled on, or at, the margin of colonies of encrusting bryozoans at a depth of ~1–3 m (Table 1 and Fig. 1E). Moreover, at depths of ~2–3 m, two recruits were observed on live shells of the bivalve *Hiatella arctica* (Table 1 and Fig. 1F), a third recruit was found on the shell plate of a barnacle (*Balanus* cf. *balanus*; Table 1 and Fig. 1G). Finally, a fourth recruit was observed on a sea urchin spine (*Strongylocentrotus droebachiensis*; not illustrated). In Kataapik and Katak sites, recruits and juveniles were only observed on the typical

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Table 1. Summary of novel and typical substrata utilised by newly settled recruits and early juveniles of *Cucumaria frondosa* in Qikiqtait, Nunavut, northern Canada.

Substratum	Number of recruits	Body length (mm)	Depth (m)	Date	Site
Novel substratum					
<i>Chorda</i> cf. <i>borealis</i>	~30	1	1	5 October 2024	Kataaluk
<i>Fucus distichus</i>	6	1–2	Tide pools	2 October 2024	Sanikiluaq
<i>Fucus distichus</i>	4	1–1.5	1	5 October 2024	Kataaluk
Coralline alga	3	1–2	1–3	5 October 2024	Kataaluk
Encrusting bryozoan	A few dozen	1–2	1–3	5 October 2024	Kataaluk
<i>Hiatella arctica</i>	2	1	2–3	25 May 2023	Kataaluk
<i>Balanus</i> cf. <i>balanus</i>	1	1	2–3	25 May 2023	Kataaluk
<i>Strongylocentrotus droebachiensis</i>	1	1.5	2–3	25 May 2023	Kataaluk
Typical and previously known substratum					
<i>Mytilus</i> sp.	340	1–24	2–3	29 April 2024	Kataapik
<i>Mytilus</i> sp.	30	1.5–19	2–3	23 May 2023	Katak
<i>Mytilus</i> sp.	12	1–2.5	2–3	25 May 2023	Kataaluk
Bare rock	~800	0.5–16.5	1–3	5 October 2024	Kataaluk
Bare rock	120	0.5–9.5	2–3	25 May 2023	Kataaluk
Bare rock	49	1–9	2–3	3 October 2024	Kataaluk
Bare rock	10	1–4	2–3	23 May 2023	Katak
Bare rock	9	1–2	Tide pools	15 November 2023	Sanikiluaq

nursery substrata of mussel shells and bare stones, down to depths of ~3 m (Table 1).

On rare occasions, recruits and early juveniles occur outside crevices and on exposed surfaces – for instance, on the blade or holdfast of seaweeds and on the plate of barnacles – which contradicts the majority of observations that were made initially (Hamel et al. 2023) and the known settlement preferences of new recruits previously described under laboratory conditions by Hamel and Mercier (1996). Further, early juveniles, aged 12 months or older (~3.6 mm or longer), preferred coralline algae (*Lithothamnion glaciale*) under experimental conditions (Gianasi et al. 2018). Presumably, a dense canopy of seaweeds provides sufficient shelter (e.g. shade) to stimulate the onset of settlement of pentactulae and to entice early juveniles to stay put instead of moving to find refuge, as is normally observed in *C. frondosa* (Hamel and Mercier 1996). Because the newly described nursery substrata (this contribution) represented only a small fraction of the available substrata, it is likely that these settlements were accidental. They may also be the result of the so called “desperate larva” theory, whereby the pentactulae might have run out of time and resources to search for more optimal or suitable microhabitat onto which to settle and recruit, as suggested for the sea cucumber *Isostichopus fuscus* (Hamel and Mercier 2022).

Although they represent only a small proportion of the new recruits or small juveniles found along the coast of Qikiqtait,

the newly described nursery substrata used by *C. frondosa* suggest that the settlement phase may be less selective than previously thought. Moreover, the use of diversified settlement substrata (especially seaweeds) by *C. frondosa* suggests that the annual recruitment pulse could be higher than previously established in the inshore subtidal waters of Qikiqtait (Hamel et al. 2023), reinforcing the notion that this area of northern Canada represents the largest known nursery of sea cucumbers on the globe.

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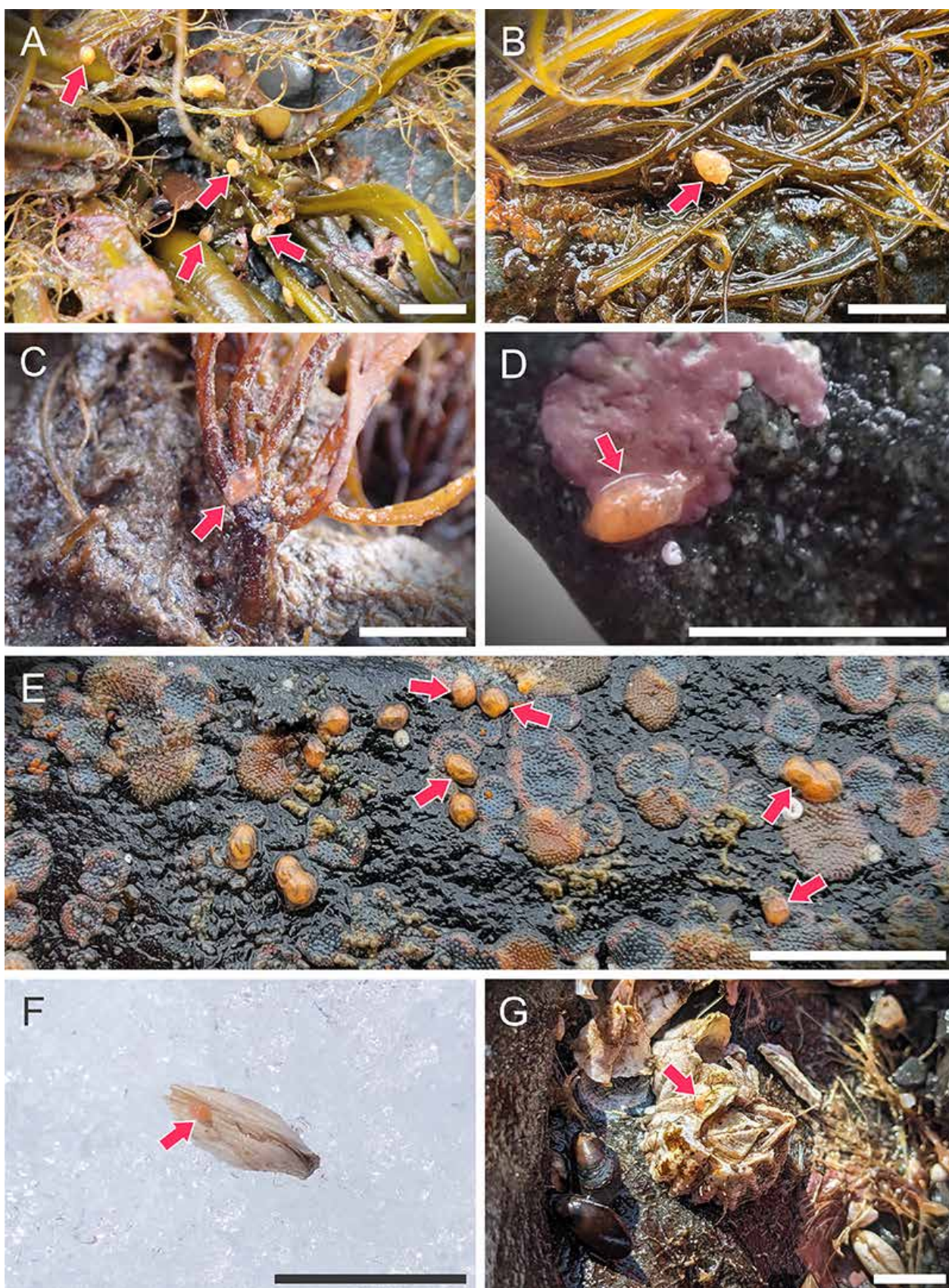


Figure 1. Newly settled recruits of the sea cucumber *Cucumaria frondosa* on A) the blades of *Chorda* cf. *borealis* and *Fucus distichus*, B) the blade of *C. cf. borealis*, C) the holdfast of *F. distichus*, D) a coralline alga, E) colonies of encrusting bryozoan, F) *Hiatella arctica*, and G) a barnacle *Balanus* cf. *balanus*. Arrows indicate juveniles on living biotic substrata. Scale bar = ~5 mm in panels A–D and ~10 mm in panels E–G. Photo credits: J.-F. Hamel (A–D); K.C.K. Ma (panels E–G).

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